

Q1 - 26 June - Shift 1

In a vernier callipers, each cm on the main scale is divided into 20 equal parts. If tenth vernier scale division coincides with ninth main scale division. Then the value of vernier constant will be $\times 10^{-2}$ mm.

*Space for your notes:***Q2 - 26 June - Shift 2**

A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is $\underline{\hspace{1cm}} \times 10^{-6}$ m.

*Space for your notes:***Q3 - 28 June - Shift 2**

A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.19 mm and 1.20 mm. The percentage error is $\frac{x}{121}\%$. The value of x is $\underline{\hspace{1cm}}$

*Space for your notes:***Q4 - 29 June - Shift 2**

Questions

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The vernier constant of Vernier callipers is 0.1 mm and it has zero error of (-0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is

Space for your notes:

5. The corrected diameter will be _____ $\times 10^{-2}$ cm.

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Answer Key

Q1 (5) **Q2 (5)** **Q3 (150)** **Q4 (180)**

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Model 1: BYC

Q1 (5)

$$20 \text{ MSD} = 1 \text{ cm}$$

$$1 \text{ MSD} = \frac{1}{20} \text{ cm}$$

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

$$= \frac{9}{10} \times \frac{1}{20} \text{ cm}$$

$$1 \text{ VSD} = \frac{9}{200} \text{ cm}$$

$$\text{VC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= \frac{1}{20} \text{ cm} - \frac{9}{200} \text{ cm}$$

$$= \frac{1}{200} \times 10 \text{ mm}$$

$$\text{VC} = 5 \times 10^{-2} \text{ mm}$$

Ans. 5

Q2 (5)

$$50 \text{ VSD} = 49 \text{ MSD}$$

$$1 \text{ VSD} = \frac{49}{50} \text{ MSD}$$

$$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= \left(1 - \frac{49}{50}\right) \text{ MSD} = \frac{1}{50} \text{ MSD}$$

$$1 \text{ MSD} = \frac{1}{40} \text{ cm}$$

$$\text{Least count} = \frac{1}{50 \times 40} \text{ cm}$$

$$= \frac{1}{2000} \text{ cm} = \frac{1}{2} \times 10^{-5} \text{ m}$$

$$= 0.5 \times 10^{-5} \text{ m}$$

$$= 5 \times 10^{-6} \text{ m}$$

Q3 (150)

$$X = \frac{1.22 \text{ mm} + 1.23 \text{ mm} + 1.19 \text{ mm} + 1.20 \text{ mm}}{4}$$

$$X = 1.21 \text{ mm}$$

$$\Delta x = \frac{0.01 + 0.02 + 0.02 + 0.01}{4} = \frac{0.06}{4} = 0.015$$

$$\text{Percentage error} = \frac{0.015}{1.21} \times 100$$

$$X = 150$$

Q4 (180)

$$\text{Measured diameter} = \text{MSR} + \text{VSR} \times \text{VC}$$

$$= 1.7 + 0.01 \times 5$$

$$= 1.75$$

$$\text{Corrected} = \text{Measured} - \text{Error}$$

$$= 1.75 - (-0.05)$$

$$= 1.80 \text{ cm}$$

$$= 180 \times 10^{-2} \text{ cm}$$

$$180$$

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